# AN ORDER ON THE LEECH LATTICE FROM TRANSPOSITION-TWISTED OCTONIONS

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Abstract. We show that the Leech lattice $\Lambda$ admits the structure of an order in a 24-dimensional real algebra. The product is built from three identical copies of a *transposition-twisted* octonion algebra — the algebra obtained by applying a single transposition to the imaginary basis elements of the standard Fano-plane multiplication — with $\mathbb{Z}_3$-symmetric cross-block routing. The construction sits at the end of a short chain of transpositions: Kirmse $\to$ Coxeter $\to$ $\sigma$-twist, each step a transposition of two basis indices that resolves a closure obstruction at the next level — Coxeter's transposition turns Kirmse's $E_8$ candidate into an order in the octonions, and the present $\sigma$-twist turns the triple-octonion product on $\Lambda$ into an order.

This result was developed through a systematic human–AI collaboration; the complete research record is publicly available [23].

## 1. Introduction

The Leech lattice $\Lambda$ is the unique even unimodular lattice of rank 24 with no vectors of squared norm 2 [4]. Its minimal shell $\mathrm{Min}(\Lambda)$ consists of 196,560 vectors of squared norm 8, and its automorphism group is the Conway group $\mathrm{Co}_0$, a double cover of the sporadic simple group $\mathrm{Co}_1$.

In rank 8, the $E_8$ lattice $L$ carries an octonionic multiplication that makes it a maximal order in the real octonion division algebra. There are seven such orders, one for each of the seven maximal orders of the integral octonions [6, 5]. Wilson [22] showed that $\Lambda$ embeds naturally in $L^3$ via three sublattice conditions, raising the question: does $\Lambda$ inherit an algebraic structure from the octonion product?

This question has been studied from several angles. Dixon [8] constructed $\Lambda$ using an octonionic XY-product. Baez and Egan [2] explored Jordan-algebraic structures on $\Lambda$ within the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$, finding closure under a doubled Jordan product.

The standard triple-octonion product on $L^3$ — using the same octonion multiplication in all three blocks with $\mathbb{Z}_3$-symmetric cross-block

*Date*: 16 April 2026.

2020 *Mathematics Subject Classification.* 11H06, 17A75, 11H56.

*Key words and phrases.* Leech lattice, octonions, non-associative order, integral algebra.

routing — satisfies two of Wilson's three sublattice conditions automatically, but closure on the third requires a block-sum sublattice $Ls$ to be preserved under octonion multiplication, and the standard product does not preserve it.

The main result of this paper is that a single transposition applied to the Fano-plane triples properly closes the remaining condition. The key insight is that the twist maps the block-sum sublattice $Ls$ to $\sigma(Ls)$, a different sublattice of $L$ that *is* closed under the standard octonion product.

**Theorem 1.1.** *Let $\cdot_\sigma$ denote the transposition-twisted octonion multiplication on $\mathbb{R}^8$ (Definition 3.1), and let $\star$ denote the associated triple product on $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ with $\mathbb{Z}_3$ cross-block routing (Definition 3.5). Then for all $u, v \in \Lambda$, the product $u \star v$ lies in $\Lambda$.*

**Corollary 1.2.** $(\Lambda, +, \star)$ *is an order in the* $\mathbb{R}$*-algebra* $(\mathbb{R}^{24}, +, \star)$.

*Proof.* $\mathrm{Min}(\Lambda)$ generates $\Lambda$ over $\mathbb{Z}$ [4]. The product $\star$ is bilinear (it is a sum of bilinear octonion products). By Theorem 1.1, $\star$ maps $\Lambda \times \Lambda$ into $\Lambda$. □

*Remark* 1.3 (Order-closure, not shell-closure). The closure statement $\Lambda \star \Lambda \subseteq \Lambda$ means that products land somewhere in $\Lambda$; it does not assert that the minimal shell is preserved. Empirically, the product norm on $\mathrm{Min}(\Lambda) \times \mathrm{Min}(\Lambda)$ takes values in $\{16, 32, 48, 64, 80, 96, 112, 128\}$ (see Section 5), none of which equals the minimal value $N = 8$: products of minimal vectors always land on strictly higher shells. This is consistent with $(\Lambda, +, \star)$ being an order in the ring-theoretic sense — a subring that is a free $\mathbb{Z}$-module of rank equal to the ambient algebra's dimension — a property weaker than, and independent of, any shell-level closure.

## 2. Preliminaries

2.1. **The octonion algebra.** Let $\mathbb{O}$ denote the real octonion algebra with basis $\{e_0, e_1, \ldots, e_7\}$, where $e_0$ is the identity. The multiplication is determined by the seven *Fano triples*

$$(1) \quad (1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3),$$

encoding the rule: for each triple $(a, b, c)$, $e_a \cdot e_b = +e_c$, with cyclic permutations carrying a positive sign and anti-cyclic permutations carrying a negative sign [1]. Every imaginary unit squares to $-e_0$. The algebra $\mathbb{O}$ is the unique real normed division algebra of dimension 8 [1].

The *norm form* is $N(x) = x\bar{x} = \sum_{i=0}^{7} x_i^2$ for $x = \sum_i x_i e_i$, where $\bar{x} = x_0 e_0 - \sum_{i=1}^{7} x_i e_i$ is the conjugate. The composition property $N(xy) = N(x)N(y)$ holds [1].

2.2. **The $E_8$ lattice.** Wilson's $E_8$ lattice is $L = D_8^+$, the set of vectors in $\mathbb{R}^8$ whose coordinates (in the basis $\{e_0, \ldots, e_7\}$) are either all integers or all half-integers, with even coordinate sum [22, 5]. It has 240 roots (vectors of squared norm 2). Under the standard octonion multiplication, $L$ is a maximal order in $\mathbb{O}$: it is closed under multiplication [6, 5].

*Remark* 2.1 (Kirmse's convention versus ours). The closure $L \cdot L \subseteq L$ depends on which Fano-plane labelling defines the octonion product. Kirmse [20] originally proposed a specific labelling under which $D_8^+$ is *not* closed — a subtle error corrected by Coxeter [6], who showed that permuting two basis indices yields a different multiplication on the same $\mathbb{R}^8$ under which $D_8^+$ *is* closed. There are seven such Coxeter-corrected conventions, each giving one of the seven maximal orders of the integral octonions [5]; see also Petersson [21] for a modern account.

The "standard" Fano triples (1) used in this paper are one of the Coxeter-corrected conventions, not Kirmse's. We have verified $L \cdot L \subseteq L$ directly under our convention: using an explicit $\mathbb{Z}$-basis of $L$ and exact rational arithmetic, all 64 basis products are integer combinations of the basis.

*Remark* 2.2 (Closure of the lattice, not the shells). When we say $L$ is closed under multiplication, we mean closure of the lattice $L$, not of any individual shell. Since the octonion product is norm-multiplicative, the product of two roots (squared norm 2) has squared norm 4 and lies on the second shell of $L$, not among the roots.

The same distinction will apply to $(\Lambda, +, \star)$ in Theorem 1.1: products of minimal Leech vectors land on higher shells of $\Lambda$, not back in the minimal shell (see Section 5).

2.3. **Wilson's Leech lattice.** Define $s = \frac{1}{2}(-e_0 + e_1 + \cdots + e_7)$ and $\bar{s} = \frac{1}{2}(-e_0 - e_1 - \cdots - e_7)$, both of norm $N(s) = N(\bar{s}) = 2$. Then $s \in L$ (it is a type-2 root, with an odd coordinate sum), while $\bar{s} \notin L$ (its coordinate sum is $-4$, which falls in the excluded even branch of the half-integer case). Let

$$L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\}, \qquad Ls = \{\ell \cdot s : \ell \in L\}.$$

Although $\bar{s}$ itself does not lie in $L$, the set $L\bar{s}$ nonetheless lies in $L$: for every $\ell \in L$ the product $\ell \cdot \bar{s}$ lies in $L$ (directly verified on the 240 roots of $L$ and extended by linearity). Both $L\bar{s}$ and $Ls$ are therefore sublattices of $L$, with $[L : L\bar{s}] = [L : Ls] = 16$, $L\bar{s} + Ls = L$, and $L\bar{s} \cap Ls = 2L$ [22].

**Definition 2.3** (Wilson [22, Section 3])**.** The *Leech lattice* is

$$\Lambda = \big\{(x, y, z) \in L^3 : (1)\ x, y, z \in L;\ (2)\ x{+}y,\ x{+}z,\ y{+}z \in L\bar{s};\ (3)\ x{+}y{+}z \in Ls\big\}.$$

The 196,560 minimal vectors of $\Lambda$ decompose into three families [22]:

| Family | Formula | Count |
|---|---|---|
| Type 1 | $(2\lambda, 0, 0)$ and cyclic permutations | $3 \times 240 = 720$ |
| Type 2 | $(\lambda\bar{s},\ (\lambda\bar{s})j,\ 0)$ and cyclic perms | $3 \times 240 \times 16 = 11{,}520$ |
| Type 3 | $((\lambda s)j,\ \lambda k,\ (\lambda j)k)$ and cyclic perms | $3 \times 240 \times 16^2 = 184{,}320$ |

Here $\lambda$ ranges over the 240 roots of $L$, and $j, k \in J = \{\pm e_t : 0 \le t \le 7\}$ (16 unit octonions).

## 3. The construction

**Definition 3.1** (Transposition-twisted octonion algebra)**.** Let $\sigma = (s\ t)$ be a transposition on $\{1, \dots, 7\}$. Extend $\sigma$ to a linear map $\sigma : \mathbb{R}^8 \to \mathbb{R}^8$ that fixes $e_0$ and permutes $e_s \leftrightarrow e_t$. Apply $\sigma$ to each standard Fano triple: $(a, b, c) \mapsto (\sigma(a), \sigma(b), \sigma(c))$. The resulting triples define a new multiplication table on $\mathbb{R}^8$. Equivalently, the *$\sigma$-twisted multiplication* is

$$x \cdot_\sigma y \;=\; \sigma\big(\sigma(x) \cdot \sigma(y)\big), \tag{2}$$

where $\cdot$ denotes the standard octonion product.

**Proposition 3.2.** *$(\mathbb{R}^8, \cdot_\sigma)$ is an octonion algebra, isomorphic to $(\mathbb{O}, \cdot)$ via the map $\sigma$ itself:*

$$\sigma(x \cdot y) = \sigma(x) \cdot_\sigma \sigma(y) \qquad \textit{for all } x, y \in \mathbb{R}^8. \tag{3}$$

*In particular, $\cdot_\sigma$ satisfies the composition property $N(x \cdot_\sigma y) = N(x)N(y)$.*

*Proof.* For any $x, y \in \mathbb{R}^8$:

$$\sigma(x) \cdot_\sigma \sigma(y) = \sigma\big(\sigma(\sigma(x)) \cdot \sigma(\sigma(y))\big) = \sigma(x \cdot y),$$

since $\sigma^2 = \mathrm{id}$. Thus $\sigma : (\mathbb{O}, \cdot) \to (\mathbb{R}^8, \cdot_\sigma)$ is a vector-space isomorphism that preserves multiplication. Composition follows. □

*Remark* 3.3. For the transposition $\sigma = (1\ 2)$, exactly 30 of the 64 multiplication table entries change relative to the standard table: 6 entries undergo a sign change only, 16 undergo a target-index change only, and 8 undergo both.

*Remark* 3.4. The 21 transpositions of $\{1, \dots, 7\}$ form a single orbit under the Fano-plane automorphism group $\mathrm{GL}(3, \mathbb{F}_2)$ (which has order 168 and acts on the 7 points of the Fano plane). Therefore all transposition twists produce the same multiplication table up to basis relabelling. The construction is essentially unique.

**Definition 3.5** (Triple product with $\mathbb{Z}_3$ routing)**.** Decompose $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ into three copies of $\mathbb{R}^8$ (blocks indexed $\alpha = 1, 2, 3$). For $u = (x, y, z)$ and $v = (x', y', z')$ with $x, x' \in \mathbb{O}_1$, etc., define the *triple product*

$$u \star v = (P_1,\ P_2,\ P_3), \tag{4}$$

where each block $P_\alpha$ is the sum of three $\sigma$-twisted octonion products:

$$P_1 = x \cdot_\sigma x' + z \cdot_\sigma y' + y \cdot_\sigma z', \tag{5}$$

$$P_2 = y \cdot_\sigma y' + x \cdot_\sigma z' + z \cdot_\sigma x', \tag{6}$$

$$P_3 = z \cdot_\sigma z' + y \cdot_\sigma x' + x \cdot_\sigma y'. \tag{7}$$

The routing rule is: a product of blocks $\alpha$ and $\beta$ lands in block $\gamma$, where $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$ for $\alpha \neq \beta$, and in block $\alpha$ when $\alpha = \beta$. All nine block-pair products use the same multiplication $\cdot_\sigma$.

## 4. Proof of closure

We prove Theorem 1.1 by showing that the twisted product preserves each of Wilson's three conditions (Definition 2.3). The proof rests on four lemmas about the interaction between the transposition $\sigma$, the maximal order $L$, and the sublattices $Ls$ and $L\bar{s}$.

4.1. **Reduction to standard products.** Since each $\sigma$-twisted product has the form $a \cdot_\sigma b = \sigma(\sigma(a) \cdot \sigma(b))$, and $\sigma$ is linear, each block $P_\alpha$ is $\sigma$ applied to a sum of standard octonion products of $\sigma$-images of the inputs.

Write $X = \sigma(x)$, $Y = \sigma(y)$, $Z = \sigma(z)$, $X' = \sigma(x')$, $Y' = \sigma(y')$, $Z' = \sigma(z')$. Then:

$$P_1 = \sigma\big(X \cdot X' + Z \cdot Y' + Y \cdot Z'\big), \tag{8}$$

$$P_2 = \sigma\big(Y \cdot Y' + X \cdot Z' + Z \cdot X'\big), \tag{9}$$

$$P_3 = \sigma\big(Z \cdot Z' + Y \cdot X' + X \cdot Y'\big). \tag{10}$$

Taking sums:

- **Block sum.** $P_1 + P_2 + P_3 = \sigma\big((X + Y + Z) \cdot (X' + Y' + Z')\big)$.
- **Pairwise sums.** For example, $P_1 + P_2 = \sigma\big(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y')\big)$, and similarly for the other two pairs.

4.2. **The four lemmas.**

**Lemma 4.1.** *$\sigma(L) = L$.*

*Proof.* The lattice $L = D_8^+$ consists of all vectors in $\mathbb{R}^8$ whose coordinates are either all integers or all half-integers, with even coordinate sum. The transposition $\sigma$ permutes two coordinates, which preserves both the integer/half-integer parity and the coordinate sum. Hence $\sigma(L) \subseteq L$. Since $\sigma$ is an involution, $\sigma(L) = L$. □

**Lemma 4.2.** *$L \cdot L \subseteq L$.*

*Proof.* $L$ is a maximal order in $\mathbb{O}$ [6, 5]. □

**Lemma 4.3.** *$L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$.*

*Proof.* Both $L$ and $\sigma(L\bar{s})$ are lattices of rank 8. It suffices to check the inclusion on $\mathbb{Z}$-bases: if $\{b_1, \ldots, b_8\}$ is a $\mathbb{Z}$-basis of $L$ and $\{c_1, \ldots, c_8\}$ is a $\mathbb{Z}$-basis of $\sigma(L\bar{s})$, then $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$ holds if and only if each of the 64 products $b_i \cdot c_j$ is an integer combination of $\{c_1, \ldots, c_8\}$. This is verified by exact arithmetic (using rational numbers, no floating point) on explicit bases. All 64 products lie in $\sigma(L\bar{s})$. □

**Lemma 4.4.** $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$.

*Proof.* As in Lemma 4.3, verified on $\mathbb{Z}$-bases by exact arithmetic: all 64 products of basis vectors of $\sigma(Ls)$ lie in $\sigma(Ls)$. □

*Remark* 4.5. The sublattice $Ls$ is *not* closed under the standard octonion product: there exist basis vectors $a, b \in Ls$ with $a \cdot b \notin Ls$. This is precisely the obstruction that the twist resolves: it moves the block-sum sublattice from $Ls$ to $\sigma(Ls)$, where closure holds by Lemma 4.4.

Note that $\sigma(Ls)$ is genuinely different from $Ls$ (not just a relabelling). For example, with $\sigma = (1\ 2)$, the element $v = (-1, 0, 1, 0, 0, 1, 0, 1)$ lies in $\sigma(Ls)$ but direct computation shows it is not an integer combination of any $\mathbb{Z}$-basis for $Ls$, so $v \notin Ls$. This distinction is what makes the twist useful: $\sigma$ is not the identity on the block-sum sublattice, so it can redirect a product into a different (and, as it turns out, closed) sublattice.

*Remark* 4.6 (Two equivalent framings). Let $\star_0$ denote the analogous triple product built from the *standard* (untwisted) octonion multiplication, i.e. with $\cdot$ replacing $\cdot_\sigma$ in every block-pair product of Definition 3.5. Applying $\sigma$ block-wise gives an algebra isomorphism $\Sigma := \sigma \oplus \sigma \oplus \sigma : (\mathbb{R}^{24}, \star) \to (\mathbb{R}^{24}, \star_0)$, since each block $P_\alpha$ of $u \star v$ is by construction (8)–(10) the image under $\sigma$ of the corresponding block of $\Sigma(u) \star_0 \Sigma(v)$. Consequently the central Theorem 1.1 has two equivalent readings:

$$\Lambda \star \Lambda \subseteq \Lambda \quad \Longleftrightarrow \quad \Sigma(\Lambda) \star_0 \Sigma(\Lambda) \subseteq \Sigma(\Lambda).$$

Neither reading asserts $\Lambda \star_0 \Lambda \subseteq \Lambda$, which fails: the standard triple product obstructs Wilson's condition (3), since it would require $Ls \cdot Ls \subseteq Ls$. The twist does not rename a closure that was already present; it identifies a distinct Leech embedding $\Sigma(\Lambda) \neq \Lambda$ — distinct because $\sigma(Ls) \neq Ls$ — on which the standard triple product closes.

### 4.3. **Condition (1): each block lies in $L$.**

**Proposition 4.7.** *For all $u, v \in \Lambda$, each block $P_\alpha$ of $u \star v$ lies in $L$.*

*Proof.* Consider $P_1 = \sigma(X \cdot X' + Z \cdot Y' + Y \cdot Z')$. Since $u \in \Lambda$, condition (1) gives $x \in L$, so $X = \sigma(x) \in L$ by Lemma 4.1. Similarly $X', Y, Y', Z, Z' \in L$. By Lemma 4.2, each product $X \cdot X'$, $Z \cdot Y'$, $Y \cdot Z'$ lies in $L$. Their sum lies in $L$ (closed under addition). Applying $\sigma$ and using Lemma 4.1, $P_1 \in L$. The same argument applies to $P_2$ and $P_3$. □

4.4. **Condition (2): pairwise sums lie in $L\bar{s}$.**

**Proposition 4.8.** *For all $u, v \in \Lambda$, the pairwise sums $P_\alpha + P_\beta$ lie in $L\bar{s}$.*

*Proof.* We show $P_1 + P_2 \in L\bar{s}$; the other two cases follow by the $\mathbb{Z}_3$ symmetry of the routing.

From (8)–(9):

$$P_1 + P_2 = \sigma\big(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y')\big).$$

Since $v \in \Lambda$, condition (2) gives $x' + z' \in L\bar{s}$, so $X' + Z' = \sigma(x' + z') \in \sigma(L\bar{s})$. Similarly $Y' + Z' \in \sigma(L\bar{s})$ and $X' + Y' \in \sigma(L\bar{s})$. Since $u \in \Lambda$, we have $X, Y, Z \in L$ (by Lemma 4.1).

By Lemma 4.3, each product $X \cdot (X' + Z')$, $Y \cdot (Y' + Z')$, $Z \cdot (X' + Y')$ lies in $\sigma(L\bar{s})$. Their sum lies in $\sigma(L\bar{s})$. Applying the outer $\sigma$:

$$P_1 + P_2 = \sigma(\text{element of } \sigma(L\bar{s})) \in \sigma(\sigma(L\bar{s})) = L\bar{s},$$

since $\sigma^2 = \mathrm{id}$. □

4.5. **Condition (3): the block sum lies in $Ls$.** This is the crux of the proof. For the standard (untwisted) product the block sum would require $Ls \cdot Ls \subseteq Ls$, which does not hold (Remark 4.5). The twist redirects the computation through $\sigma(Ls)$, where closure holds.

**Proposition 4.9.** *For all $u, v \in \Lambda$, the block sum $P_1 + P_2 + P_3$ lies in $Ls$.*

*Proof.* From (8)–(10):

$$P_1 + P_2 + P_3 = \sigma\big((X + Y + Z) \cdot (X' + Y' + Z')\big).$$

Since $u \in \Lambda$, condition (3) gives $x + y + z \in Ls$, so

$$X + Y + Z = \sigma(x + y + z) \in \sigma(Ls).$$

Similarly $X' + Y' + Z' \in \sigma(Ls)$.

By Lemma 4.4, $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$, so

$$(X + Y + Z) \cdot (X' + Y' + Z') \in \sigma(Ls).$$

Applying the outer $\sigma$:

$$P_1 + P_2 + P_3 \in \sigma(\sigma(Ls)) = Ls.$$ □

This completes the proof of Theorem 1.1.

## 5. Algebraic properties

The order $(\Lambda, +, \star)$ has been tested for several standard algebraic properties. The results are summarised below.

| Property | Holds? | Details |
|---|---|---|
| Multiplicative identity | No | the ambient algebra $(\mathbb{R}^{24}, +, \star)$ has no identity element |
| Commutativity | No | $< 0.1\%$ of pairs commute |
| Norm multiplicativity | No | $\approx 47\%$ of pairs satisfy $N(u \star v) = N(u)N(v)$ |
| Left alternativity | No | $< 0.1\%$ of pairs satisfy $(x \star x) \star y = x \star (x \star y)$ |
| Right alternativity | No | $< 0.1\%$ of pairs satisfy $(x \star y) \star y = x \star (y \star y)$ |
| Flexibility | No | $\approx 8.6\%$ of pairs satisfy $(x \star y) \star x = x \star (y \star x)$ |
| Power-associativity | No | $\approx 29\%$ of elements satisfy $x^2 \star x = x \star x^2$ |

*Remark* 5.1. The product norm on $\mathrm{Min}(\Lambda) \times \mathrm{Min}(\Lambda)$ takes values in $\{16, 32, 48, 64, 80, 96, 112, 128\}$, all multiples of 16. The mode is 64 (the norm-multiplicative value $N(u)N(v) = 8 \times 8$).

*Remark* 5.2. The ambient algebra $(\mathbb{R}^{24}, +, \star)$ has no multiplicative identity: any putative left identity $(a, b, c)$ would have to act on an arbitrary $(x', y', z')$ by the block matrix $\left(\begin{smallmatrix} a & c & b \\ c & b & a \\ b & a & c \end{smallmatrix}\right)$ (entries acting by left-multiplication in $\mathbb{O}$), and no triple $(a, b, c)$ makes this the identity on all three blocks simultaneously. In particular, $(e_0, e_0, e_0)$ is not an identity: it maps $(x', y', z')$ to $(x'+y'+z',\ x'+y'+z',\ x'+y'+z')$. Consequently $(\Lambda, +, \star)$ is a non-unital order.

The order does not satisfy any of the classical algebraic identities (associativity, alternativity, flexibility, power-associativity, commutativity). The defining property is *closure*: the Leech lattice is stable under the product. The relationship between $\star$ and the Conway group $\mathrm{Co}_0$ (the automorphism group of $\Lambda$ as a lattice) is an open question.

## 6. Research methodology

This research was conducted as a collaboration between a human researcher and an AI agent (Claude, Anthropic) over a series of directed prompts, governed by a written methodology (the "Manifesto") established at the outset.

The human researcher (the author) brought the initial hypothesis — motivated by prior work on Okubo algebras and autotopies of normed composition algebras — established the research methodology, and maintained control of the inquiry through prompt-by-prompt review. The AI agent performed the computational work: implementing octonion algebras, Wilson's Leech lattice construction, trial algebras, and closure tests, and contributed mathematical analysis.

The computational verification was a vehicle in the discovery process, designed to mediate distrust into received knowledge and intermediate results. Tests were written from first principles to independently verify every foundation: that octonion multiplication satisfies the composition property $N(xy) = N(x)N(y)$, that Wilson's construction produces

a lattice with the correct minimal shell of 196,560 vectors, and that the $E_8$ lattice is indeed a maximal order. This independent verification is how we know that the properties stated in the literature are correctly implemented. Closure of the twisted product was confirmed on over 12,000,000 random pairs of minimal vectors with zero failures before the symbolic proof was found.

The symbolic proof (Section 4) and the exact-arithmetic verifications of Lemmas 4.3 and 4.4 use Python's `fractions.Fraction` type, operating entirely in $\mathbb{Q}$ with no floating-point arithmetic. The $\mathbb{Z}$-basis checks involve solving 64 systems of 8 linear equations each, all yielding integer solutions.

The methodology required:

- Every claim backed by a reference or computation.
- Every prompt logged, with actions recorded.
- Every conclusion independently verified.
- Forward-evolving corrections: no rewriting history; errors are corrected in subsequent entries.

The path to the result was not direct. Six trial algebra families were explored and ruled out before the transposition twist was identified in trial 007. The systematic methodology — understanding the anatomy of failure (which conditions fail, for which vector types, and why) before trying the next approach — was essential. The failure analysis of the untwisted product (Wilson's condition (3) per Definition 2.3, exclusively type-3×type-3) directly pointed to the mechanism that the twist needed to correct.

Iterative convergence toward a robust result — including learning along the way what it is we are actually doing — is how study, research, and discovery have always proceeded. The AI agent violated its own logging protocol on three occasions (prompts 011, 018–019, and 029–031), each time caught by the human researcher on review. Conversely, the author learned from the AI agent about errors and misconceptions in the framing of certain prompts, which likewise are part of the record. What is novel here is the ability to trace the entire path, beginning to end, through the complete list of prompts.

The full record — all prompts, code, evidence, and git history — is available at the repository [23]. The reader can form their own assessment of what this means for questions of discovery and attribution.

## 7. Related work

**Wilson (2009)** [22] characterises $\Lambda$ as $\{(x, y, z) \in L^3 : \text{conditions 1–3}\}$, gives an explicit listing of the 196,560 minimal vectors in the form $3 \times 240 \times (1 + 16 + 16 \times 16)$, and proves that $\Lambda$ is even and self-dual. The sublattice framework is the setting used in the present paper.

**Dixon (2010)** [8] constructs $\Lambda$ using the octonion XY-product[1]

$$A \circ_{XY} B \;=\; (AX)(Y^{\dagger}B),$$

where $X, Y \in \mathbb{O}$ are chosen so that $YX^{\dagger}$ is a unit octonion (the identity of the new multiplication). When $Y^{\dagger} = X^{-1}$ the XY-product specialises to the X-product $A \circ_X B = (AX)(X^{-1}B)$; Dixon uses this X-product with $X = \ell_0 = \frac{1}{2}(1 + e_1 + \cdots + e_7)$ to obtain $E_8$, and general XY-products for higher-dimensional laminated lattices including $\Lambda$.

**Baez and Egan (2014)** [2] study closure properties of integer sublattices of the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$. In parts 9–10 of the series, Baez considers the 27-dimensional lattice $\mathbb{Z}^3 \oplus L_L \subset \mathfrak{h}_3(\mathbb{O})$, with integers on the diagonal and an off-diagonal Leech sublattice $L_L \cong \Lambda$ embedded as an octonionic triple. This lattice is *not* closed under the Jordan product $x \circ y = xy + yx$, but *is* closed under the doubled product $2(xy + yx)$. Egan further enumerated at least 17,280 distinct off-diagonal Leech embeddings $L_L \subset \mathbb{O}^3$ for which the associated 27-dimensional lattice is closed under this doubled product. Baez left open the question whether any such lattice is closed under the undoubled Jordan product — which would remove the factor-of-2 discrepancy — and conjectures that the answer is negative.

**Earlier precursors.** Further back, several constructions foreshadow the octonion-triple and Jordan-algebra frameworks above. Conway and Sloane [3] give 23 constructions of $\Lambda$, one for each Niemeier lattice, including a Turyn-style construction from three copies of $E_8$. Lepowsky and Meurman [17] describe $\Lambda$ as a sublattice of $E_8^3$ containing $2E_8^3$ and use this to generate the Conway group $\mathrm{Co}_0$ via permutations of the three $E_8$ factors together with one extra automorphism. Elkies and Gross [12] obtain $\Lambda$ from an integer form of the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$ as the orthogonal complement of a distinguished identity element, and show that the automorphism group of this integer form is generated by octave reflections. Each of these constructions is additive or geometric: none defines or tests a bilinear product on $\Lambda$.

**Kirmse (1924)** [20] exhibited the $E_8$ lattice as a unimodular integral quadratic lattice inside the real octonion algebra. His lattice is not closed under multiplication, an error corrected by Coxeter [6] by swapping two basis indices in the Fano-plane labelling.

The present construction continues the same conceptual line. The Kirmse–Coxeter step modifies the octonion multiplication (a single transposition of basis indices) so that the $E_8$ lattice becomes an order. The present paper applies a second transposition of basis indices

[1]The construction has an earlier precursor in Dixon [7], where a closely related octonionic description of $\Lambda$ is proposed. The formula there required a small correction in the third family of minimal vectors, which Wilson [22, end of §5] supplied in a footnote-style remark while independently deriving the sublattice characterisation. The situation is similar to the Kirmse–Coxeter relationship discussed below: an early, structurally correct proposal needing correction.

— the twist $\sigma$ in Definition 3.1 — to the already-corrected octonion product, so that the triple-product construction on the Leech lattice becomes an order. The chain is Kirmse $\to$ Coxeter $\to$ $\sigma$-twist: each step is a transposition, and each resolves a closure obstruction at the next level of the construction.

## 8. Conclusion

The Leech lattice admits an order under a remarkably simple construction: three identical copies of a transposition-twisted octonion algebra with $\mathbb{Z}_3$ routing. The symbolic proof (Section 4) reduces closure to four lemmas, of which the crux is Lemma 4.4: the sublattice $\sigma(Ls)$ — unlike $Ls$ itself — is closed under the standard octonion product. The twist maps the condition-3 sublattice to this closed sublattice, and the involutory nature of $\sigma$ maps the result back.

Several questions remain immediately open:

- The automorphism group of $(\Lambda, +, \star)$ and its relationship to the Conway group $\mathrm{Co}_0$.
- Whether the order is maximal in any appropriate sense.
- Classification of the algebra $(\mathbb{R}^{24}, +, \star)$: it is non-associative, non-alternative, non-flexible, and non-unital (as an order), placing it outside the classical families.
- Whether other permutations of the Fano triples (beyond transpositions) also yield closed products on $\Lambda$.

## 9. Outlook

The intention in publishing this result at this stage is to invite research rather than to sit on the ideas. If others arrive at deeper answers faster, so much the better. What follows is an outlook that names the directions we find most promising and the sources that shaped the intuition leading to the present construction.

9.1. **A ternary reformulation.** A more natural classification of $(\mathbb{R}^{24}, +, \star)$ may emerge from a *ternary* rather than binary viewpoint. The Leech lattice possesses a prominent $\mathbb{Z}_3$-symmetry (the cyclic permutation of the three octonion blocks used throughout this paper), and the binary product $\star$ is assembled from nine bilinear building blocks whose routing is itself governed by that $\mathbb{Z}_3$-action. Ternary composition algebras, as studied by Elduque [9, 10, 11], are typically non-associative, non-commutative, non-alternative, and non-power-associative — the same negative classification our order satisfies — yet carry high internal symmetry, with order-3 automorphisms playing the central role: the Okubo algebra has automorphism group SU(3), and the Okubo product itself is built from an order-3 automorphism of the octonions. Complementary perspectives on the Okubo algebra and triality include the quasigroup-theoretic treatment of Smith and Vojtěchovský [19],

Kamiya and Okubo's study of algebras satisfying triality and $S_4$-symmetry [15], and Okubo's seminal book [18]. A deeper precursor is Hall's monograph [14], which establishes that Moufang loops and groups with triality are essentially the same object, giving a structural scaffolding for any triality-based reformulation. Recasting $\star$ as a trace-like reduction of a ternary operation compatible with the $\mathbb{Z}_3$-block symmetry is a natural direction for further work.

9.2. **Consequences for algebraic approaches to physics.** A classification of the ambient algebra $(\mathbb{R}^{24}, +, \star)$ matters beyond pure algebra. Programmes that seek to model fundamental physics on algebraic structures — especially those that use the four normed division algebras, triality, and related symmetric composition algebras — have grown considerably in recent years. The author previously invited [16] for research into autotopies of normed composition algebras as a mathematical setting for this line of inquiry. Recent work of Furey and Hughes [13] isolates triality itself as the operative structural skeleton: the Standard Model's $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ gauge symmetries are embedded inside $\mathfrak{tri}(\mathbb{C}) \oplus \mathfrak{tri}(\mathbb{H}) \oplus \mathfrak{tri}(\mathbb{O})$, and the triality skeleton is shown to accommodate a variety of particle-family representations — the authors' own assignment of three Standard Model generations to a triality triple in $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ is one concrete realization of that wider scaffolding.

9.3. **On the form of this work.** This paper is the outcome of a human–AI collaboration conducted entirely through a sequence of AI prompts (Section 6), with the complete record of prompts, code, intermediate results, and corrections preserved verbatim [23]. The form itself is part of the contribution. Collaborations of this kind will become more common, and this work is offered in its present form partly in the hope that the process — how a research question was framed, how the AI was directed and corrected, how false trails were recognised and closed — is instructive for mathematicians considering AI-assisted research. Comments on both substance and method are welcome.

## Acknowledgments

The author thanks Geoffrey Dixon for helpful discussions and advice on octonionic structures associated with the Leech lattice.

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